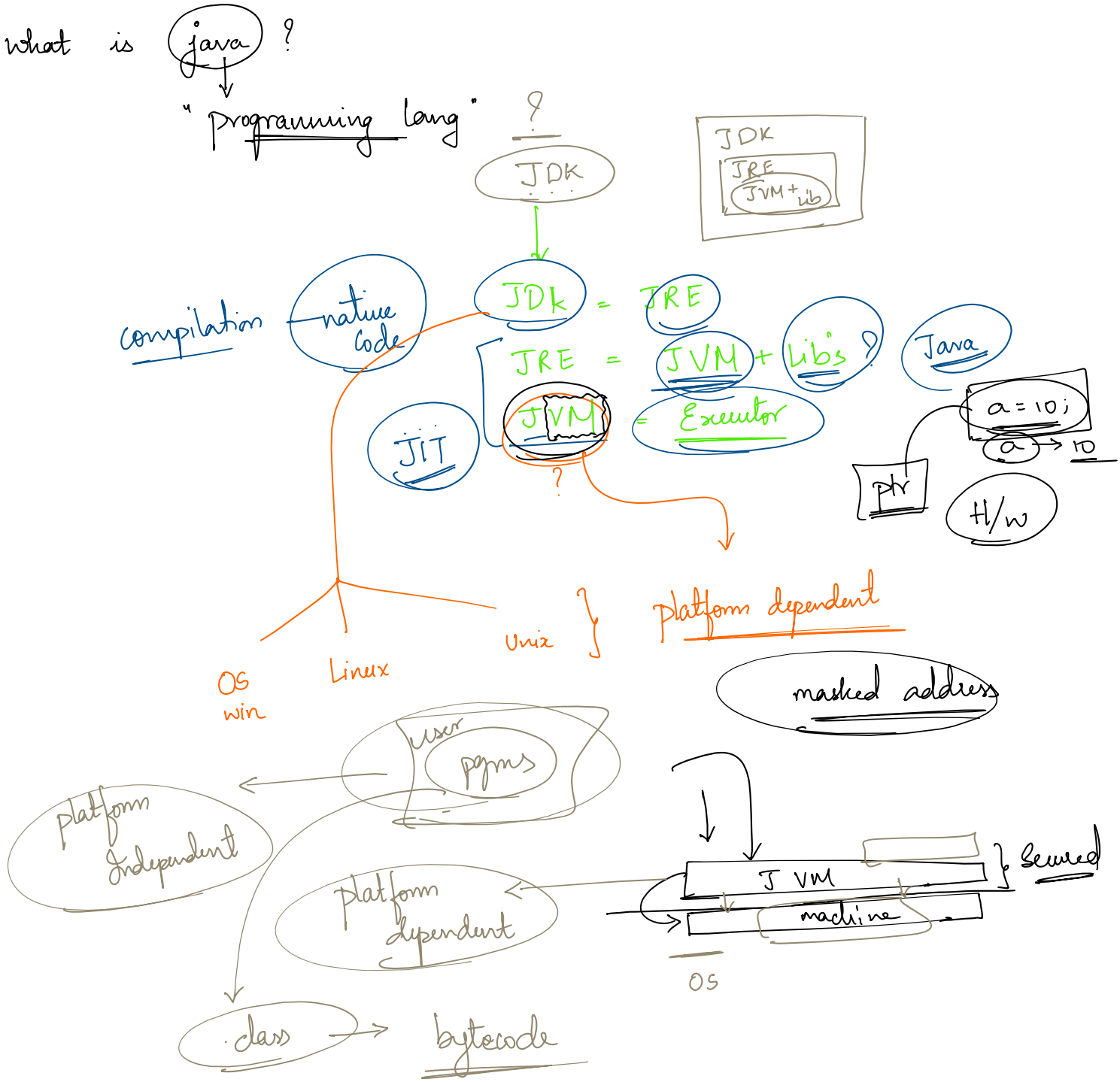
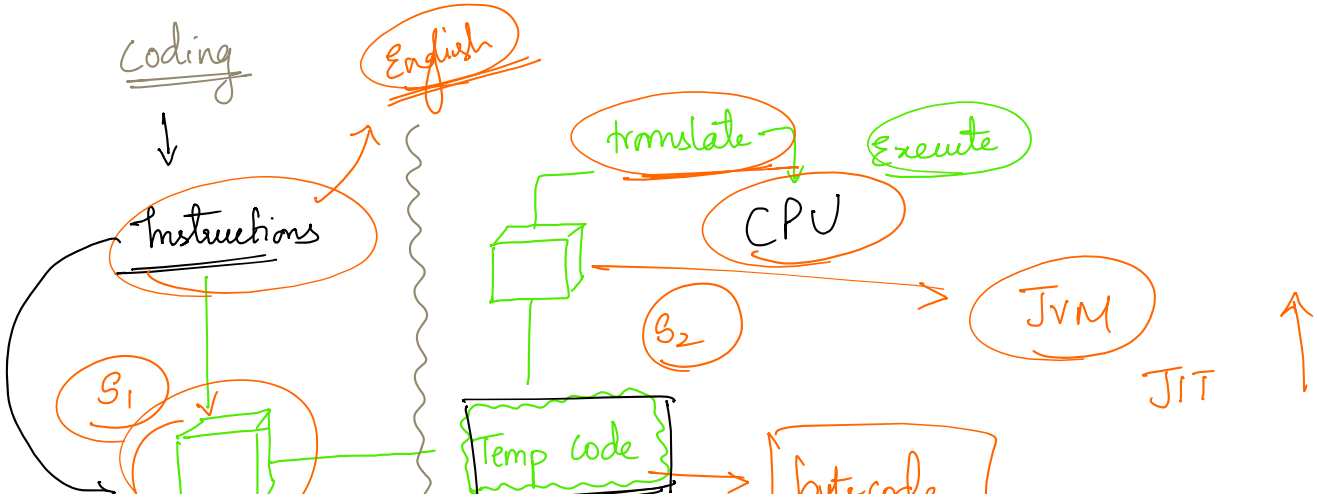
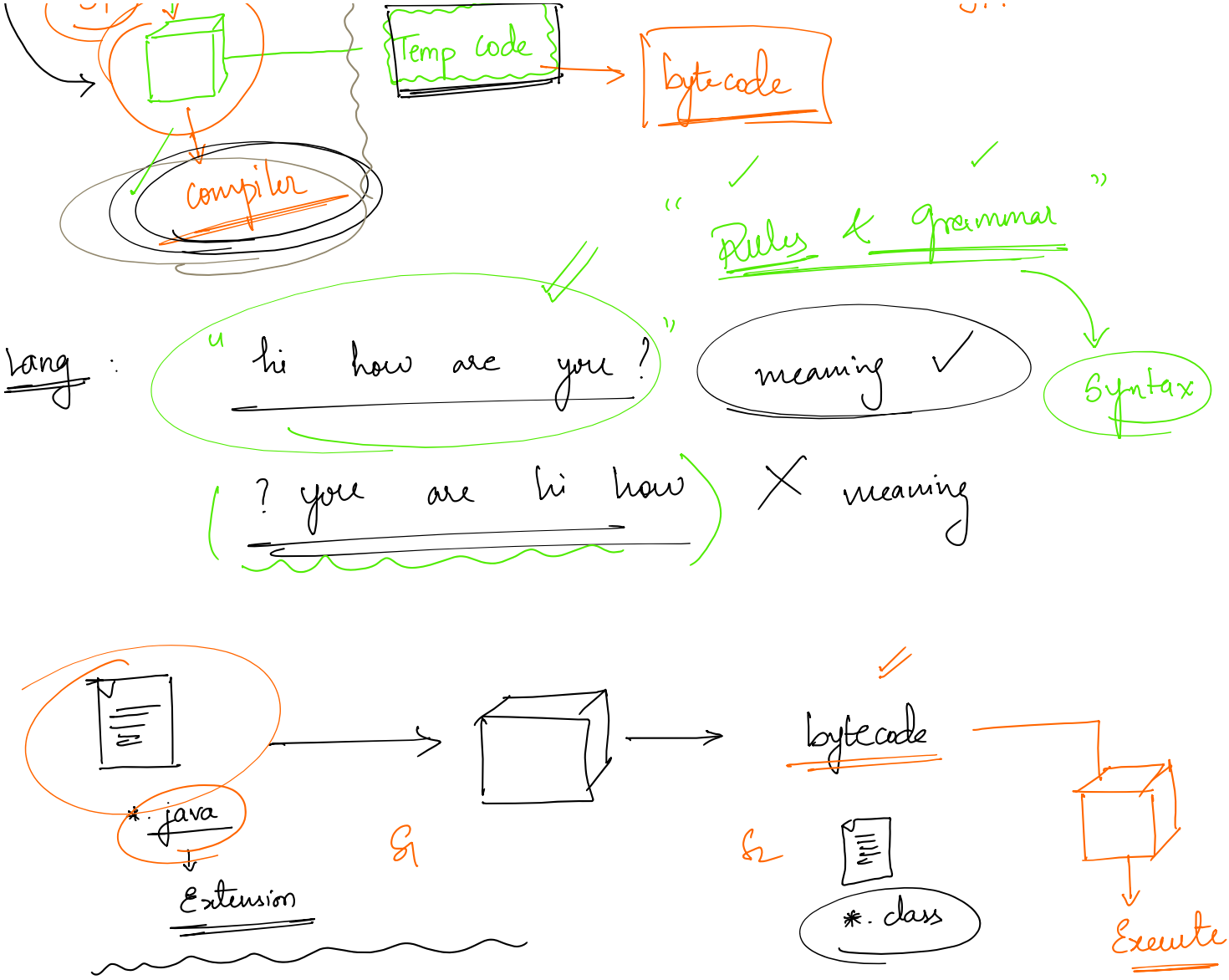

what is java ?
" programming lang "
?
JDK
JRE
JVM + Lib
JDK = JRE
JRE = JVM + Libs ?
JVM = Executor
JIT
compilation
native code
Java
a = 10;
a → 10
ptr
H/W
OS
win
Linux
Unix
platform dependent
masked address
user
pgms
platform Independent
platform dependent
J VM
machine
Secured
OS
.class
bytecode


Coding
English
instructions
translate
Execute
CPU
JVM
JIT
S1
S2
Temp code



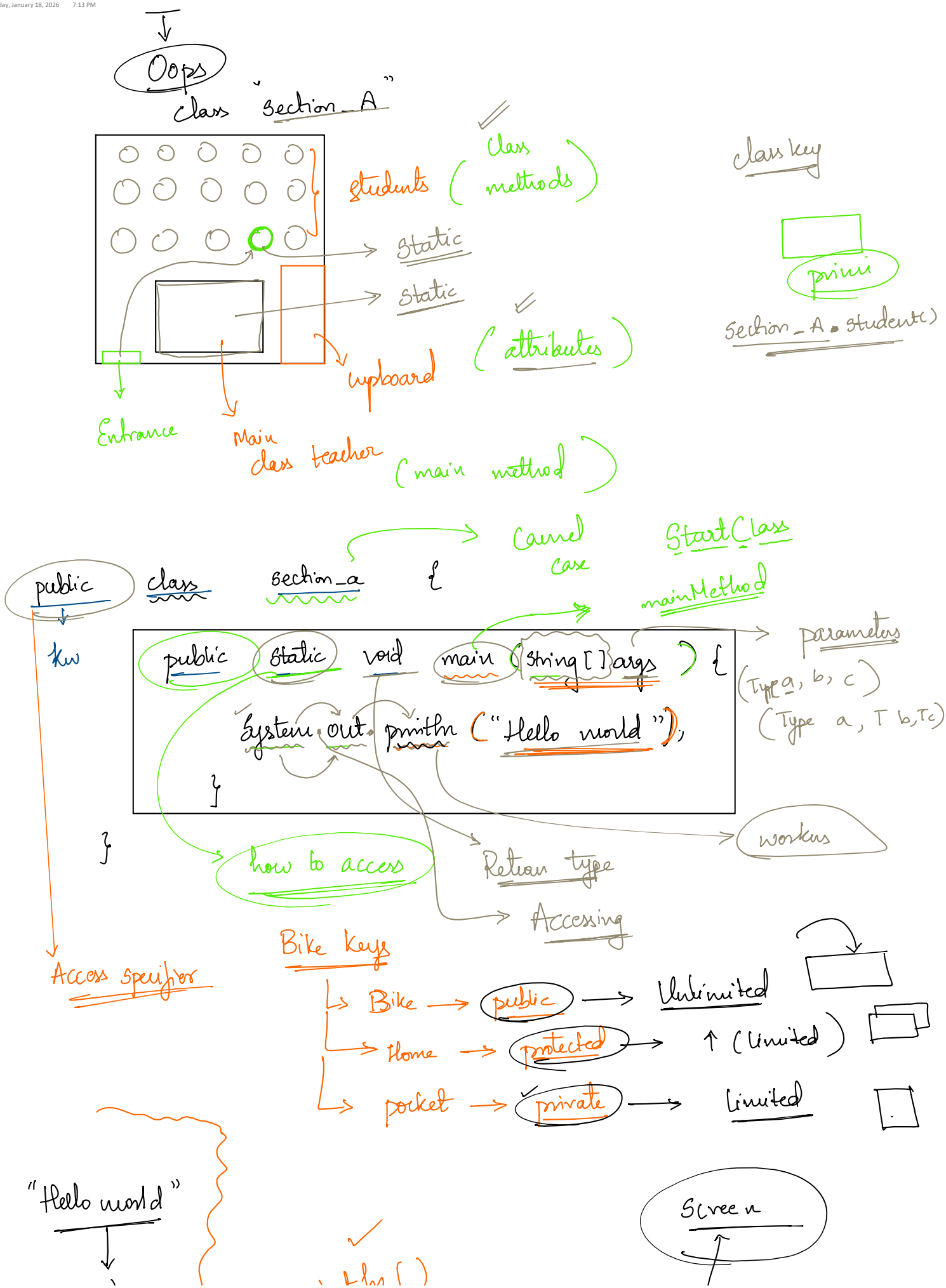
abc | abc (.....)

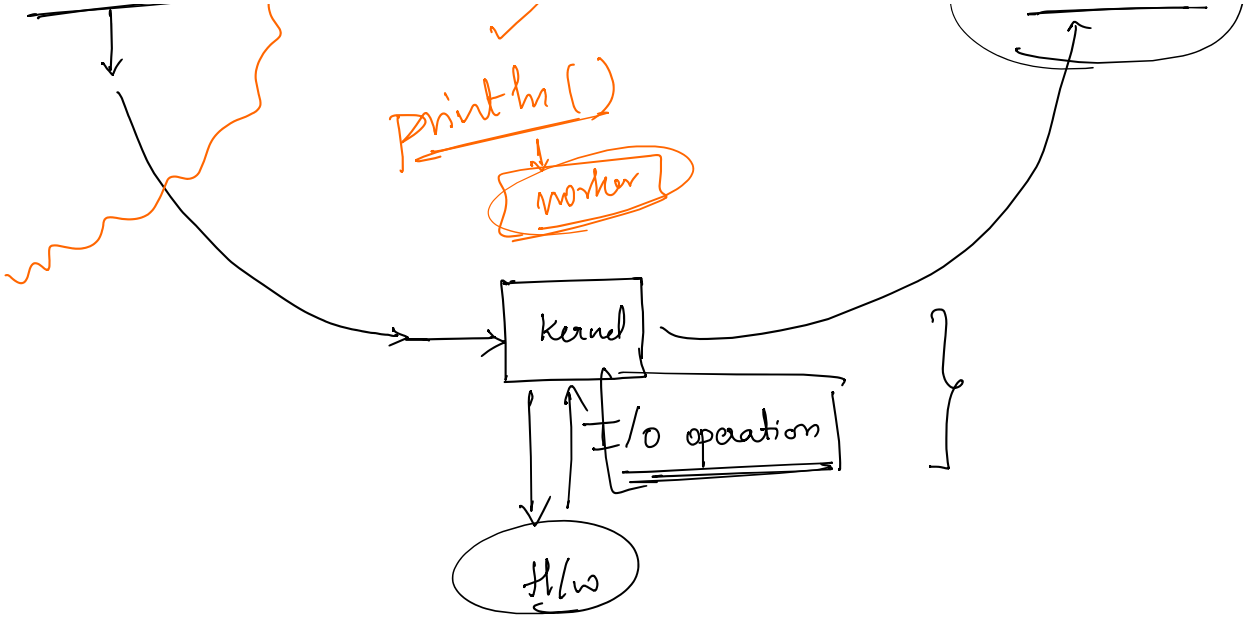
keyword function

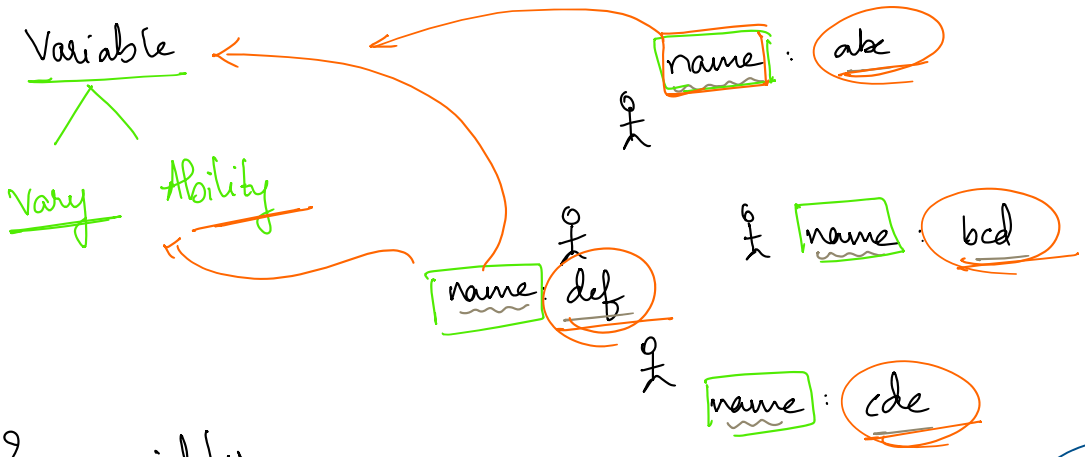
```
public class helloWorld {
    public static void main (String [] args ) {
        System. out. println ("Hello world");
    }
}
```

Understanding Structure of JAVA program

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println()
worker
Kernel
I/O operation
H/w



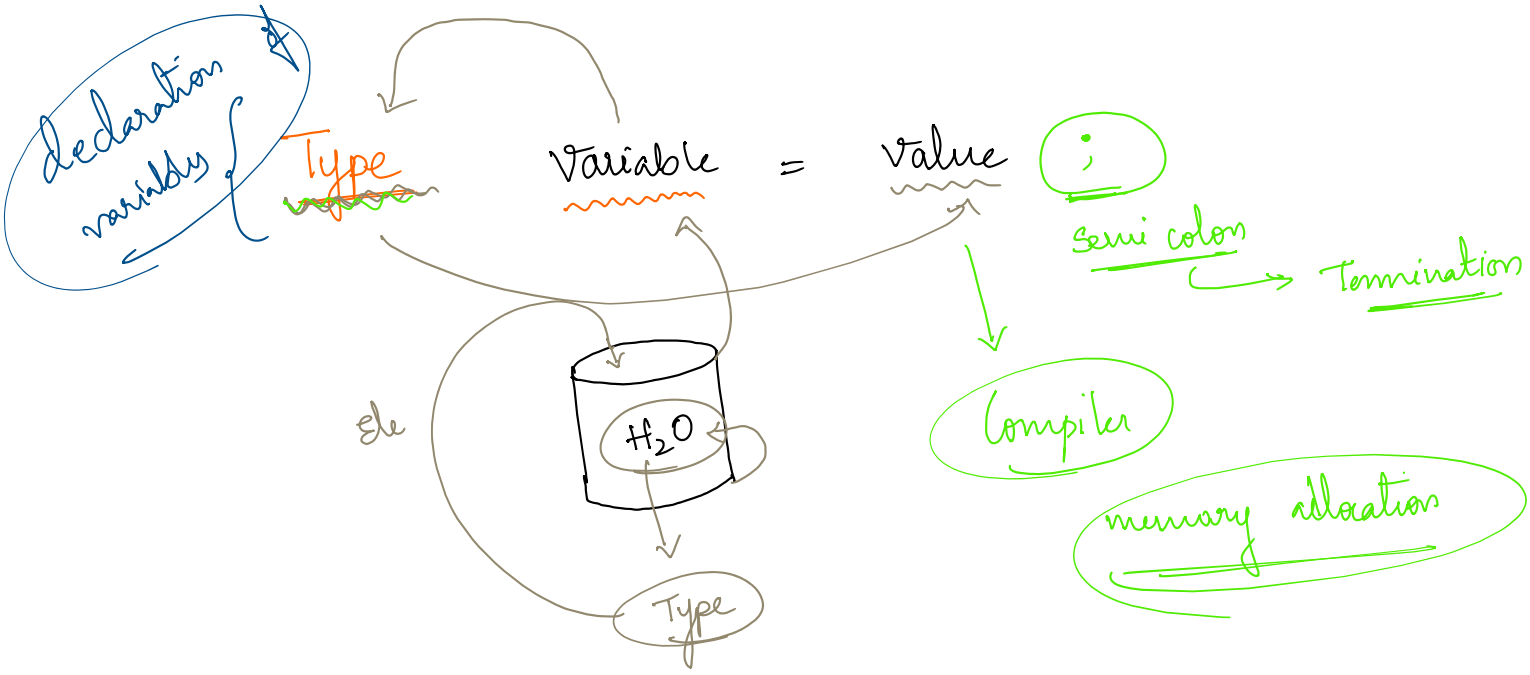
? variables

price = 130010

130010

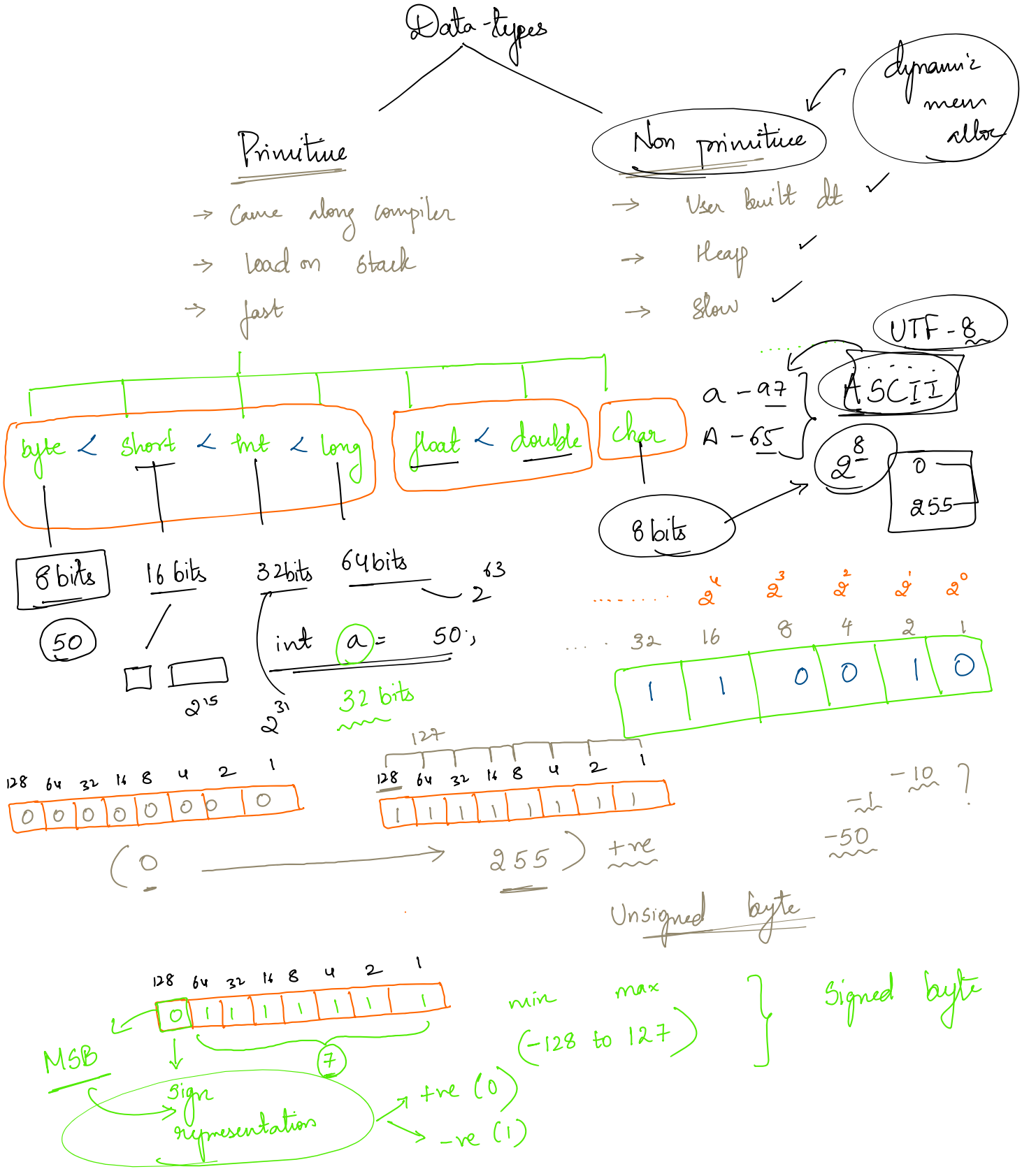
1,30,000

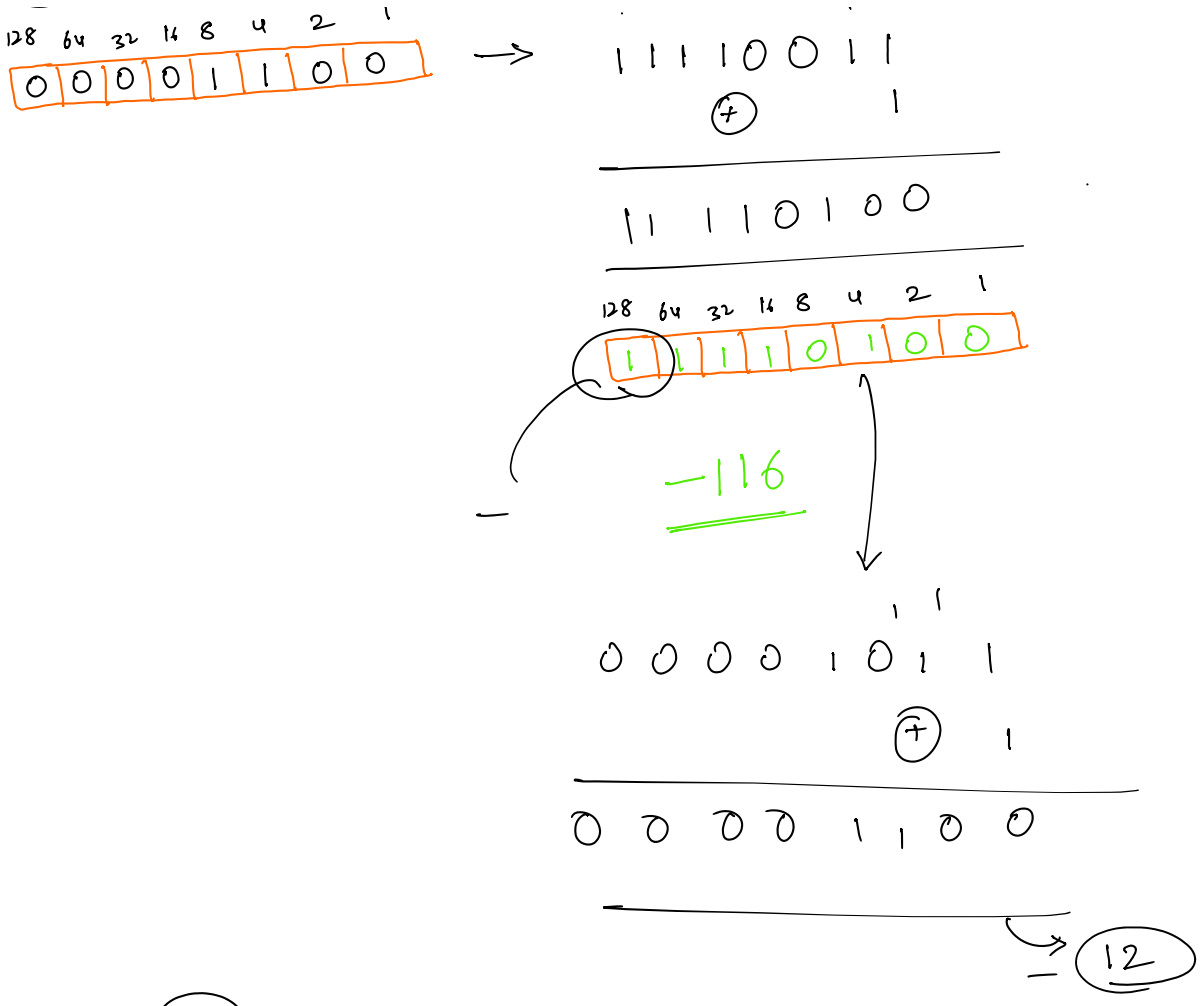
&ysout(" Gold rate in hyderabad rupees ");

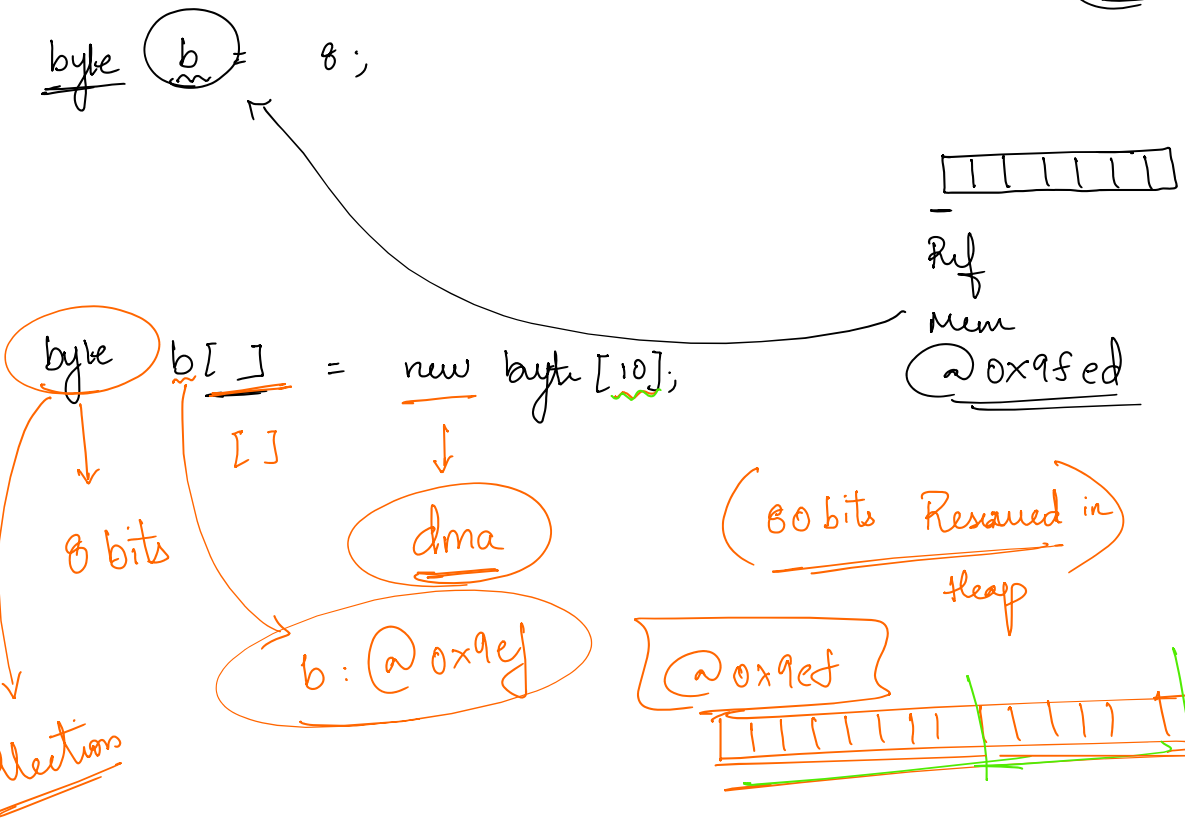


(Type var = val;)

var = val; ✓ Re-define

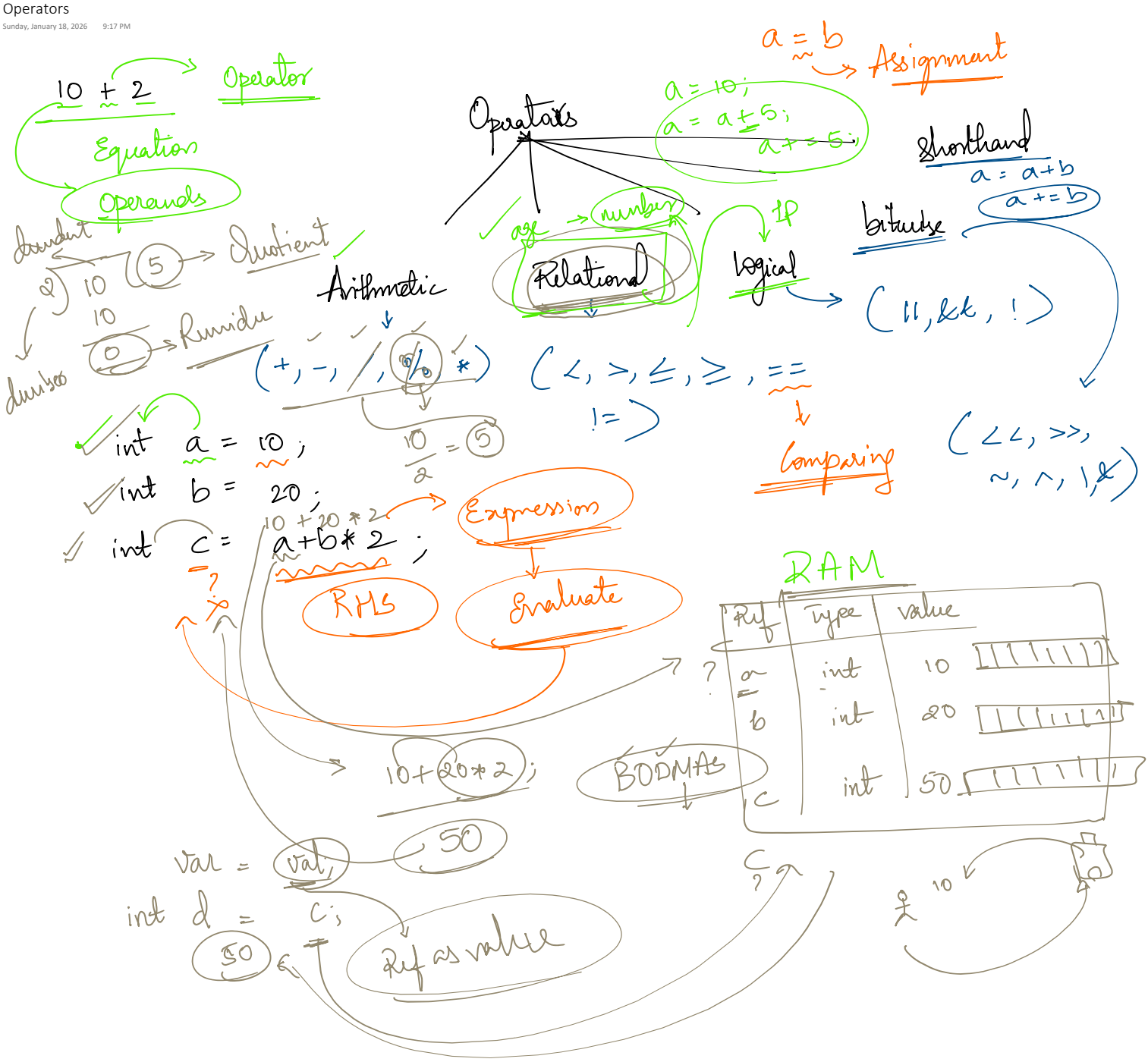

Data-types
Primitive
→ Came along compiler
→ Load on stack
→ fast
Non primitive
→ User built dt
→ Heap
→ Slow
dynamic mem allo
byte < short < int < long
float < double
char
8 bits
16 bits
32 bits
64 bits
50
int a = 50;
2^15
2^31
63
2
a - 97
A - 65
ASCII
UTF-8
2^8
0
255
8 bits
32 bits
2^4 2^3 2^2 2^1 2^0
32 16 8 4 2 1
1 1 0 0 1 0
128 64 32 16 8 4 2 1
0 0 0 0 0 0 0 0
127
1 1 1 1 1 1 1 1
(0 → 255) +ve
Unsigned byte
-10 ?
-1
-50
MSB
7
Sign representation
+ve (0)
-ve (1)
min max
(-128 to 127)
Signed byte
12
0 0 0 0 1 1 0 0
→
1 1 1 1 0 0 1 1

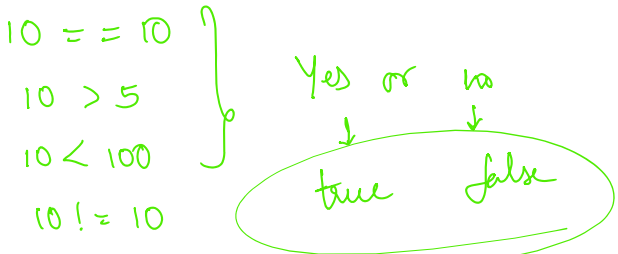

128 64 32 16 8 4 2 1
0 0 0 0 1 1 0 0
1 1 1 1 0 0 1 1
+ 1
1 1 1 0 1 0 0
128 64 32 16 8 4 2 1
1 1 1 1 0 1 0 0
-116
0 0 0 0 1 0 1 1
+ 1
0 0 0 0 1 1 0 0
12

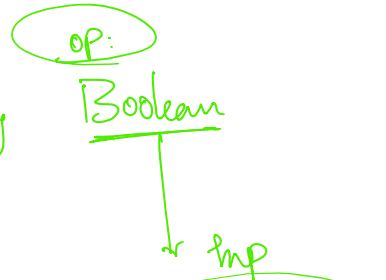

byte b = 8;
Ref
Mem
@0x9fed
byte b[ ] = new byte[10];
[ ]
8 bits
dma
b : @0x9ef
80 bits Reserved in heap
@0x9ef
Collection
80

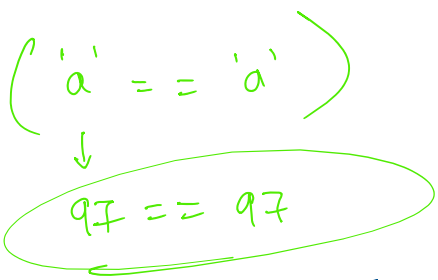
Operators

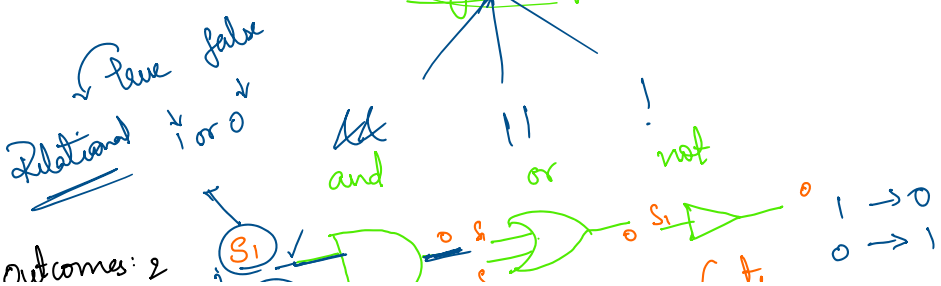
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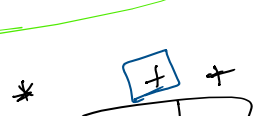


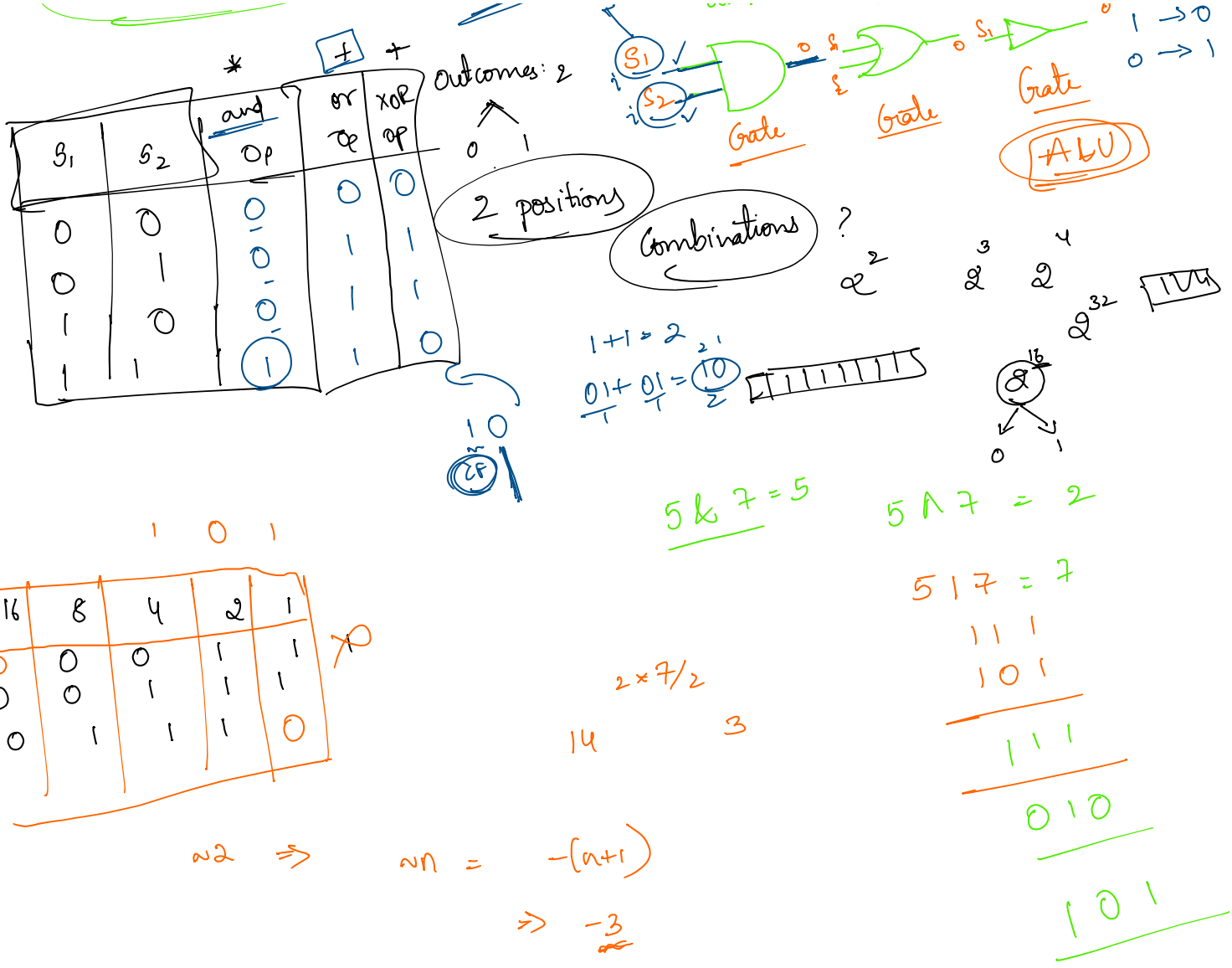






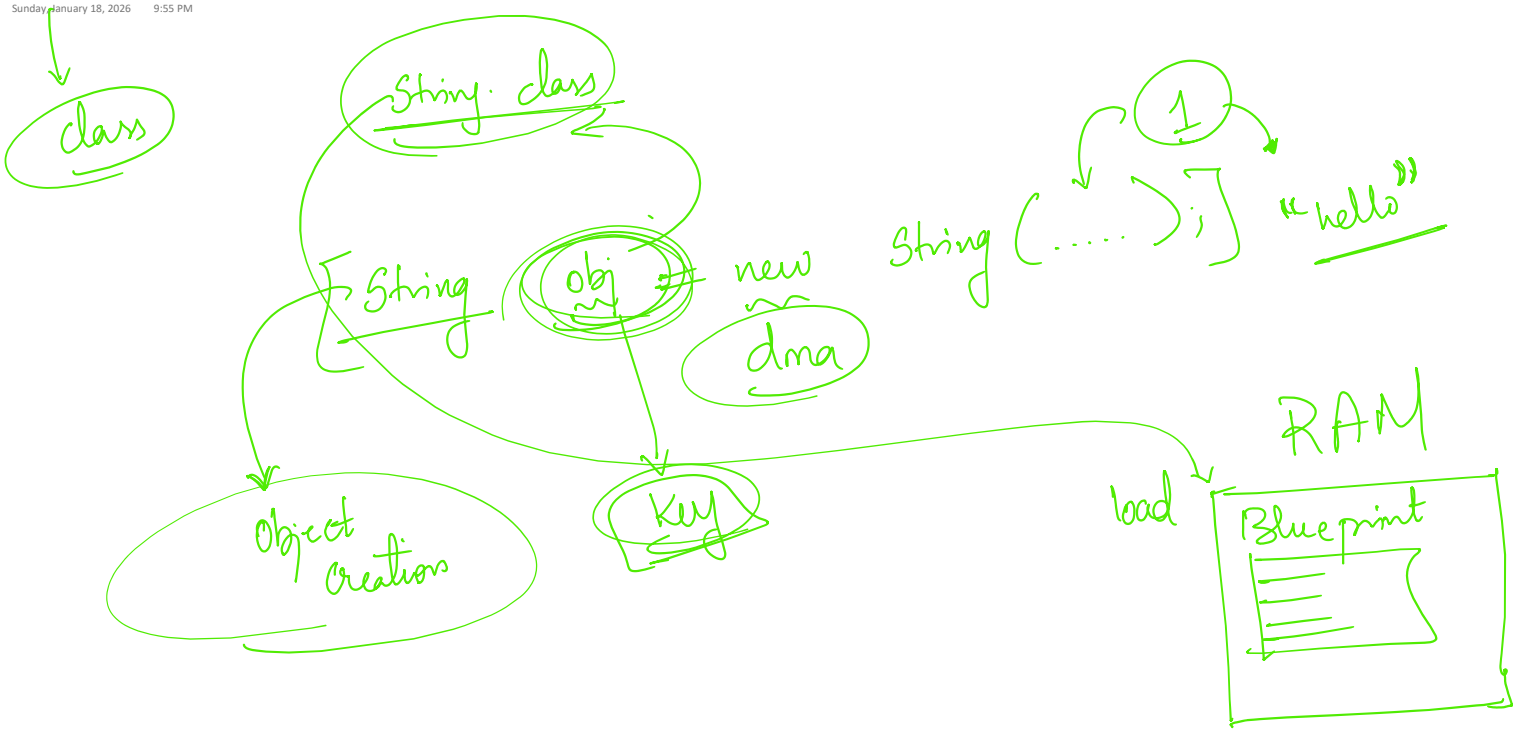





*
+
+
S_1 S_2
and op
or op
xor op
Outcomes: 2
0 1
2 positions
Combinations ?
2^2
2^3 2^4
2^{32}
1 0
CF
1+1 = 2
01 + 01 = 10
S_1
S_2
Gate
Gate
Gate
ALU
1 → 0
0 → 1
16 8 4 2 1
1 0 1
5 & 7 = 5
5 ^ 7 = 2
5 | 7 = 7
1 1 1
1 0 1
1 1 1
0 1 0
1 0 1
2 × 7/2
14
3
n2 ⇒ nn = −(n+1)
⇒ −3

Strings

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Arrays

Monday, January 19, 2026 1:31 AM

what is an array ? → Non primitive

Contiguous allocation of homogenous elements

[1, 2, 3, 4, 5, 6 . . . . .]

Similar data types

byte arr[] = new byte[5];

40 bits

8 8 8 8 8

arr → @0x9ef } Heap

int a = 10;

int a[] = {10};

arr[0] → arr[@0x9ef + (8 * 0)]

[1]

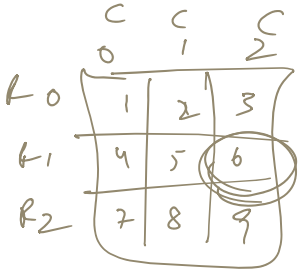
[2]

8 bits

2

byte arr[] = {1, 2, 3, 4, 5};

byte arr[] = new byte[]{1,2,3,4,5};



R x C

(R, C)

(1, 2)

[ε1, ε2, ε3]

[ ] [ ] [ ]

C

i 0 1 2

i [0] [ε1 ε2 ε3],

i 1 [ε4 ε5 ε6],

i 2 [ε7 ε8 ε9]]

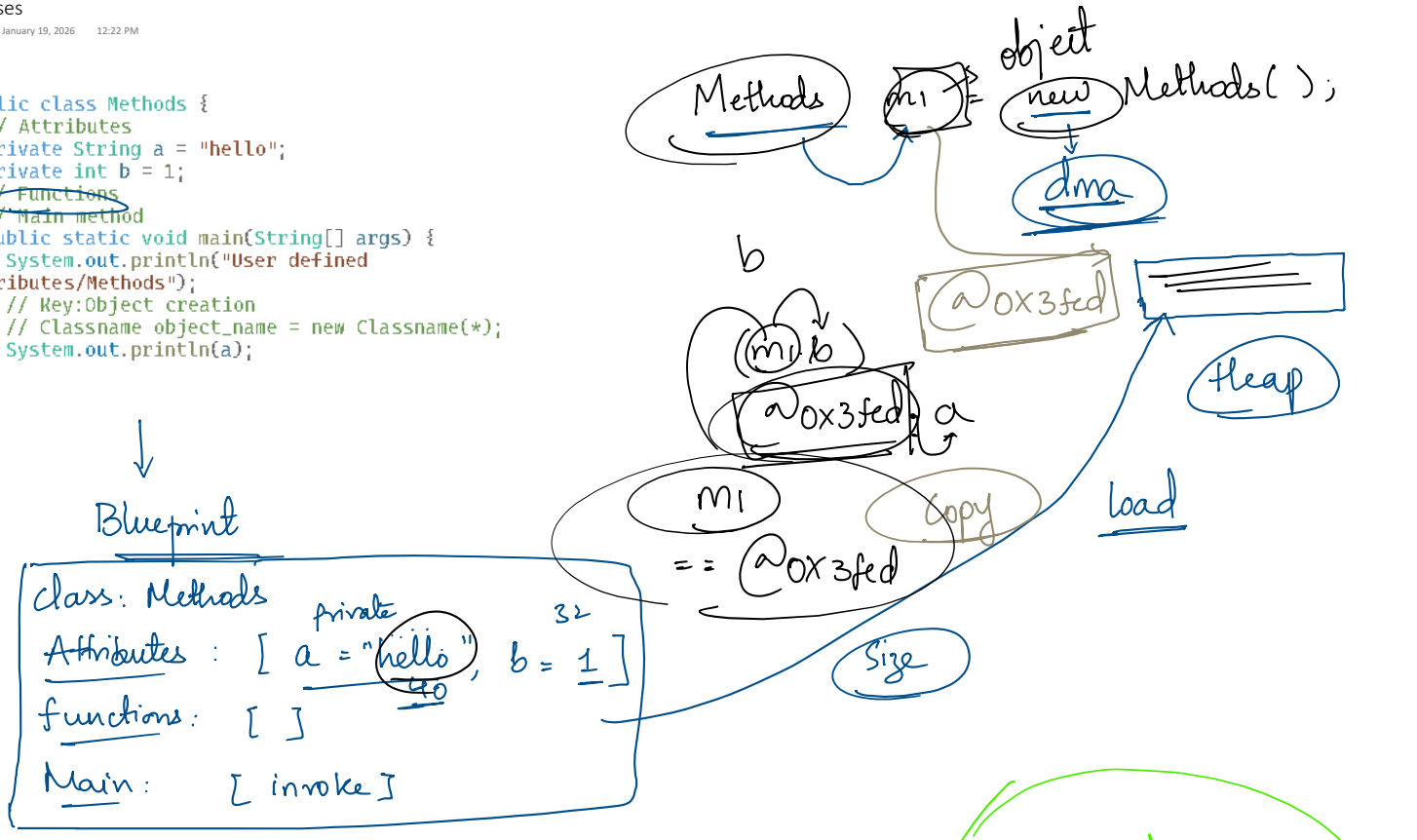
[1][2]

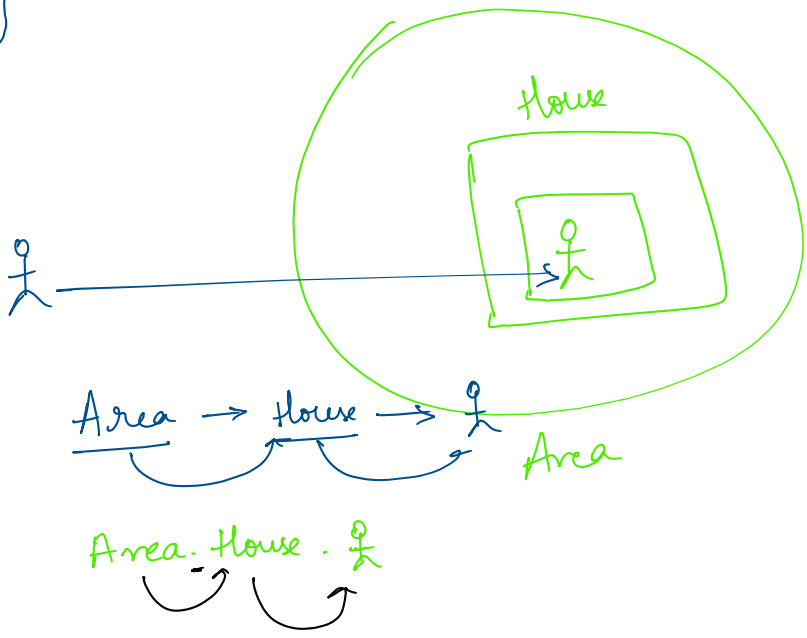
(1,2)

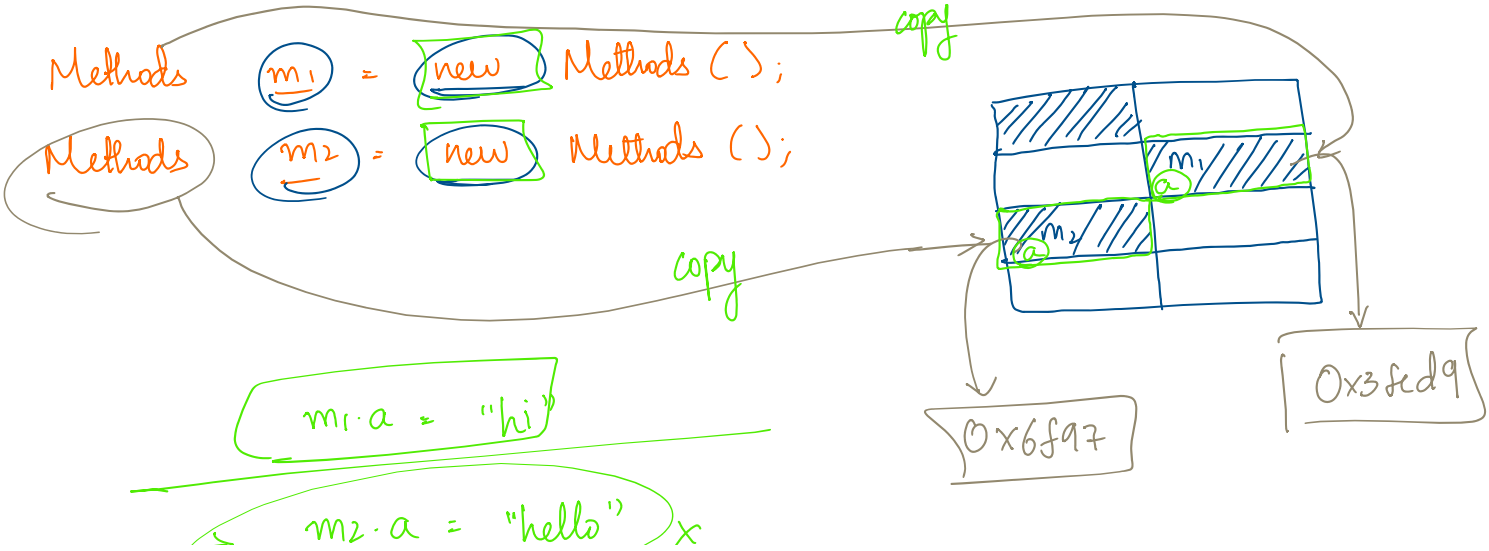
Classes

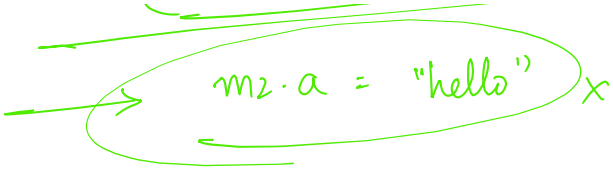
Monday, January 19, 2026 12:22 PM

```java
public class Methods {
  // Attributes
  private String a = "hello";
  private int b = 1;
  // Functions
  // Main method
  public static void main(String[] args) {
    System.out.println("User defined
Attributes/Methods");
    // Key:Object creation
    // Classname object_name = new Classname(*);
    System.out.println(a);
  }
}
```



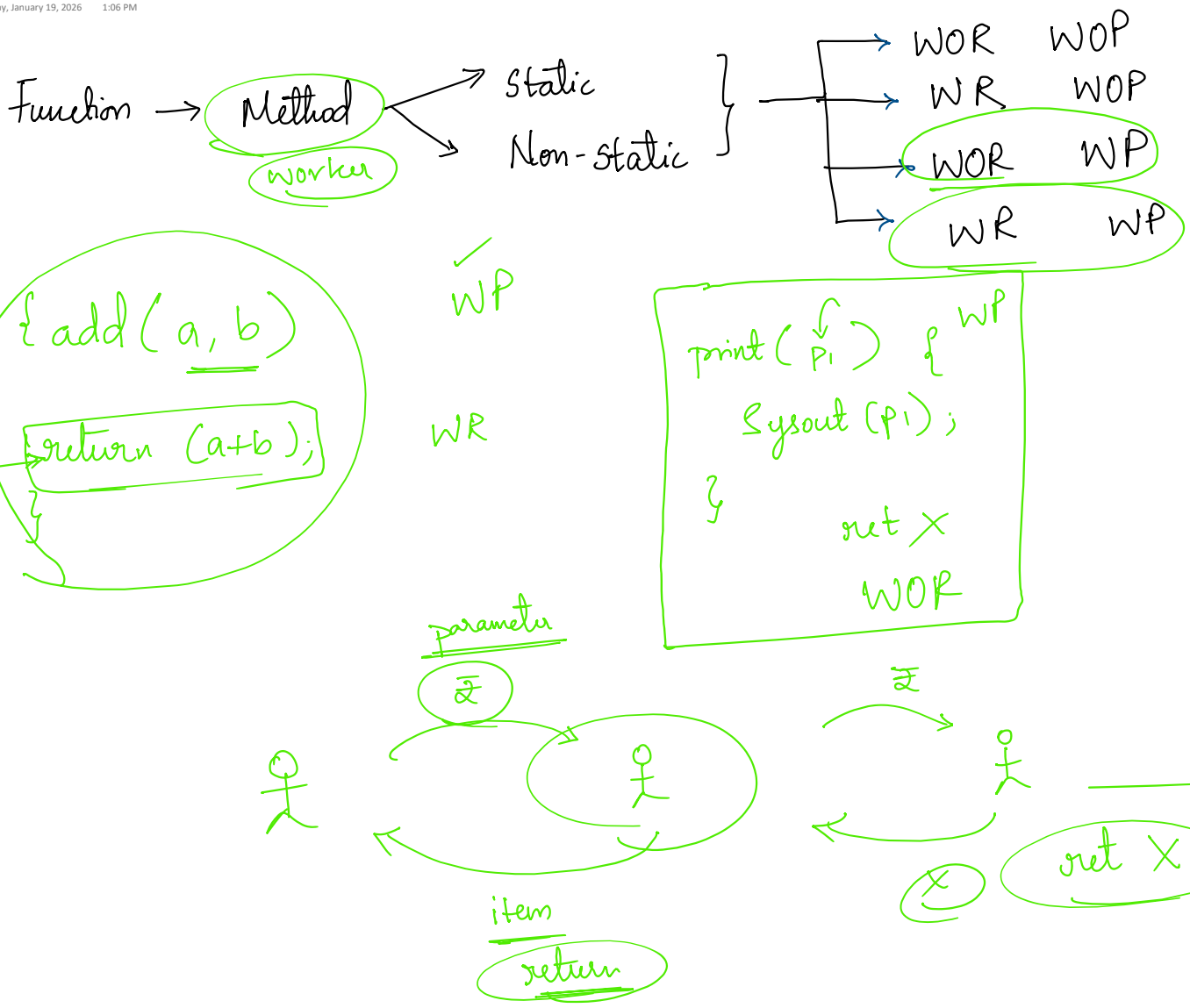





m2.a = "hello"

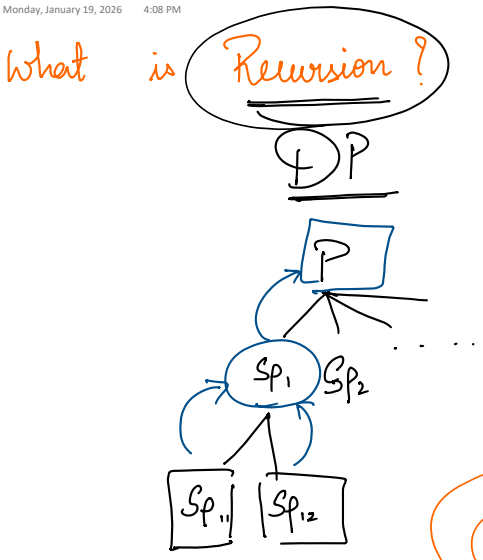
Class Functions

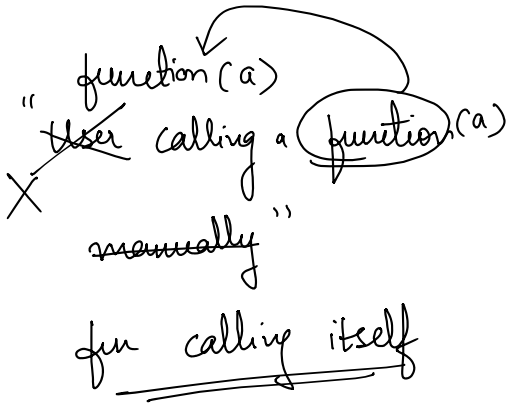
Monday, January 19, 2026 1:06 PM

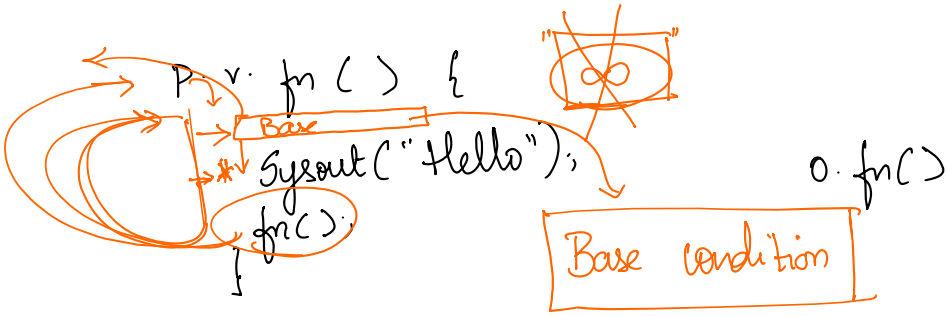


Recursion

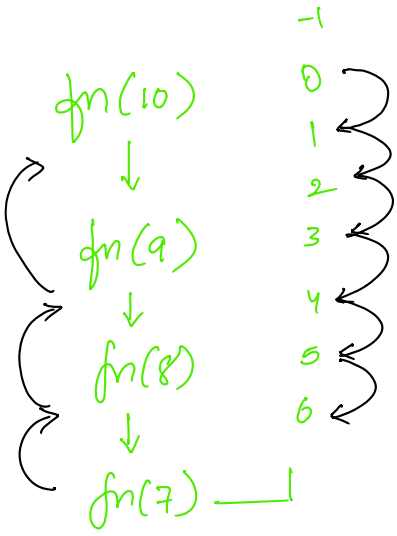
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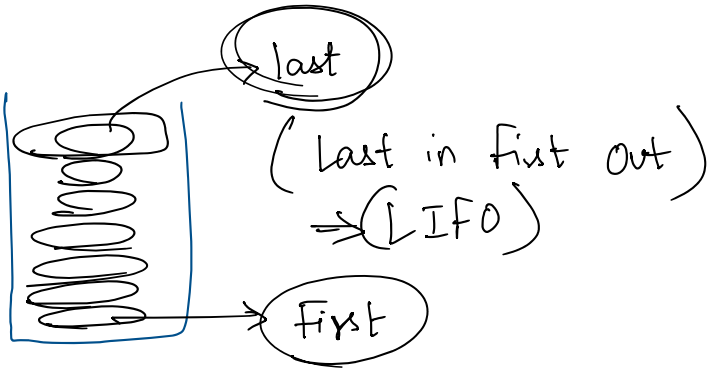


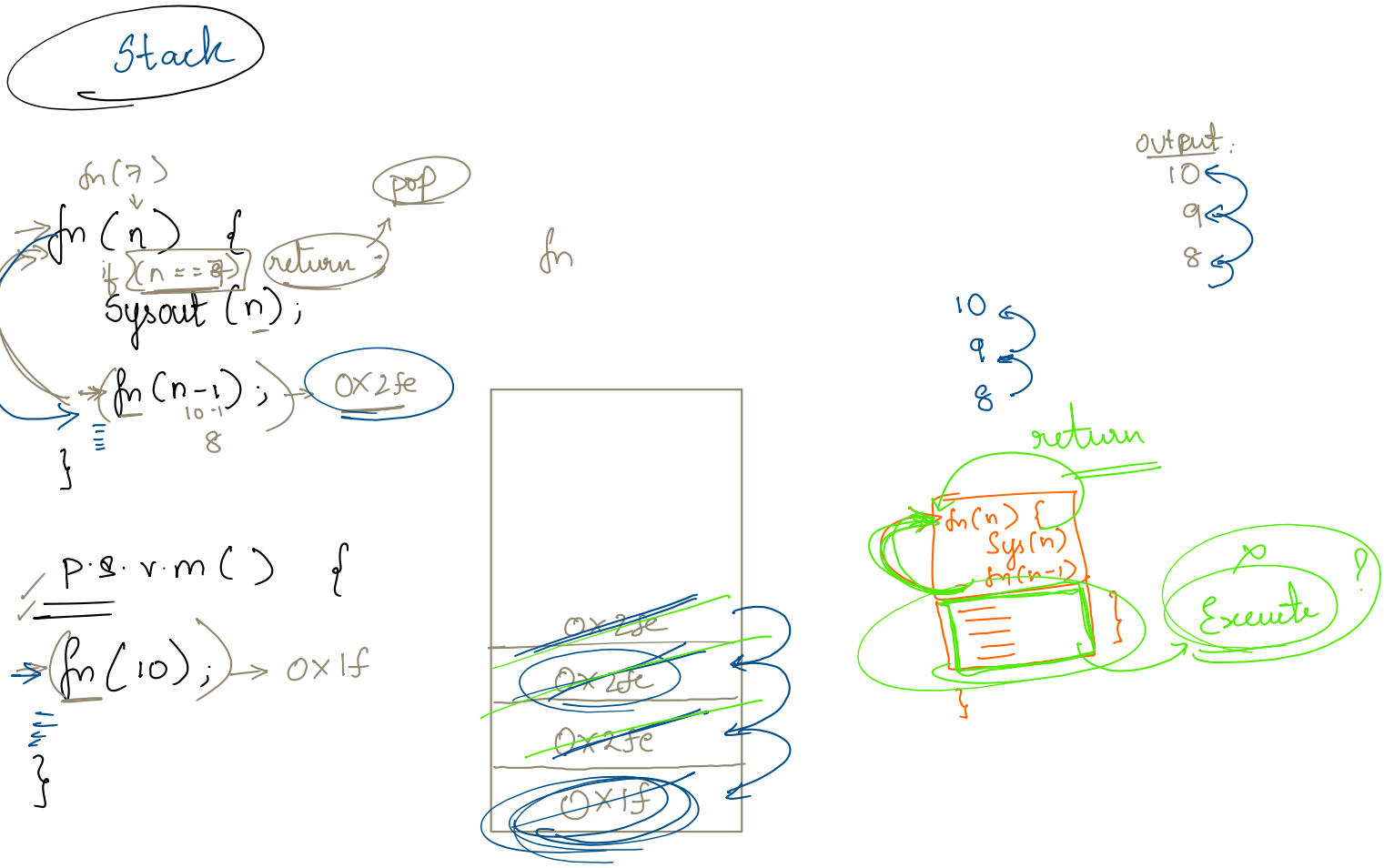

```
public void fn(n) {
// Base condition
if (n < 0) {
    return;
}
// Recursion
fn(n-1);
}
```



finite / Terminate / Halt

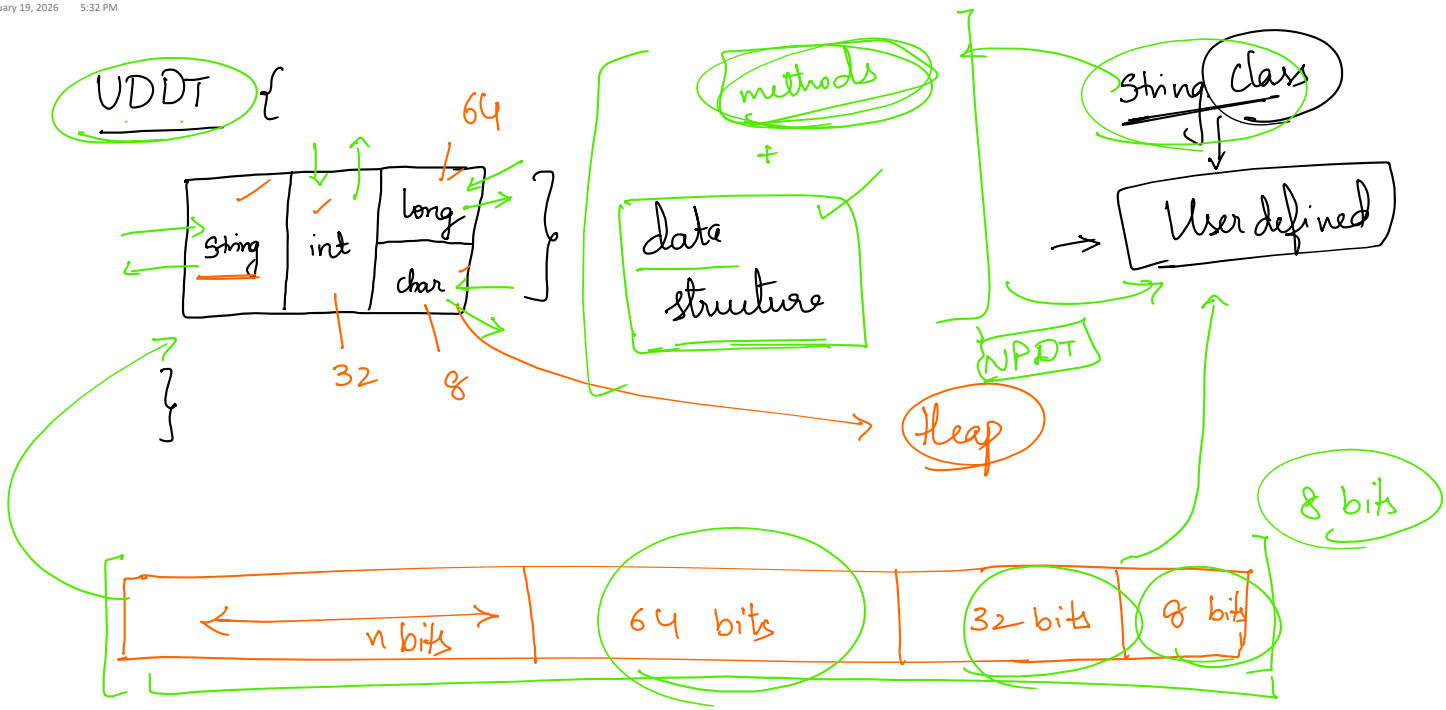
Recursion




Stack
fn(7)
fn(n) {
if (n == 8)
return
pop
Sysout (n);
fn(n-1);
10-1
8
0X2fe
}
p.s.v.m() {
fn(10);
0X1f
}
fn
0X2fe
0X2fe
0X2fe
0X1f
output:
10
9
8
10
9
8
return
fn(n) {
Sys(n)
fn(n-1)
}
Execute

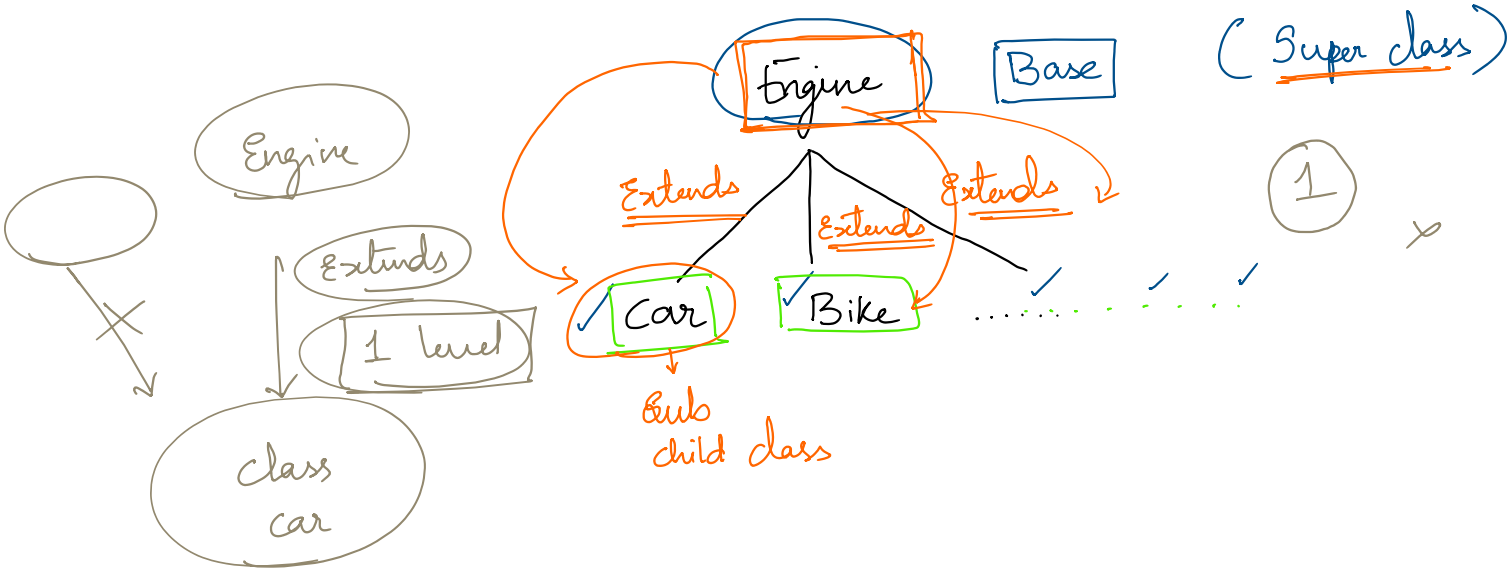
User Defined Datatype

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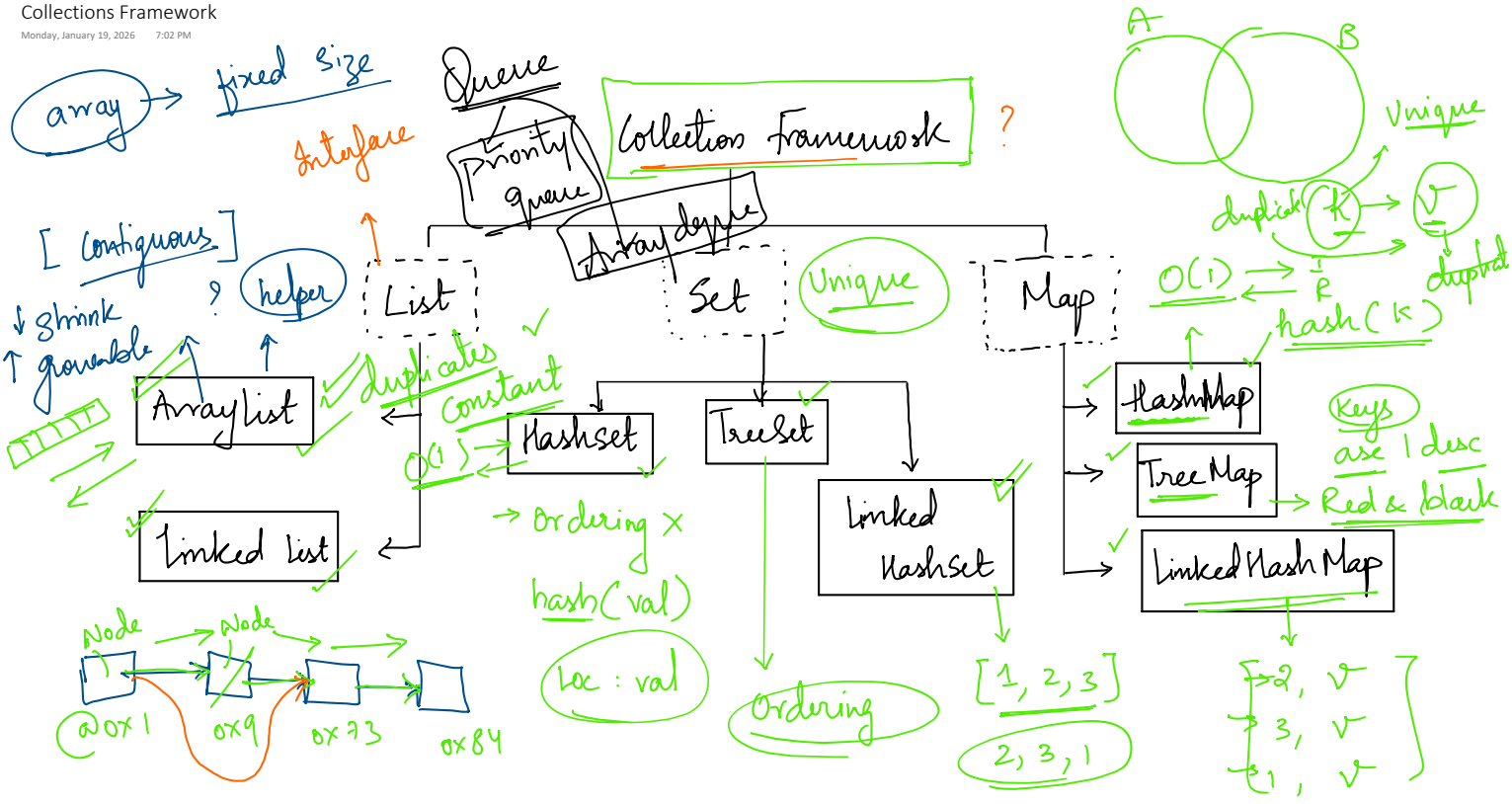
Inheritance

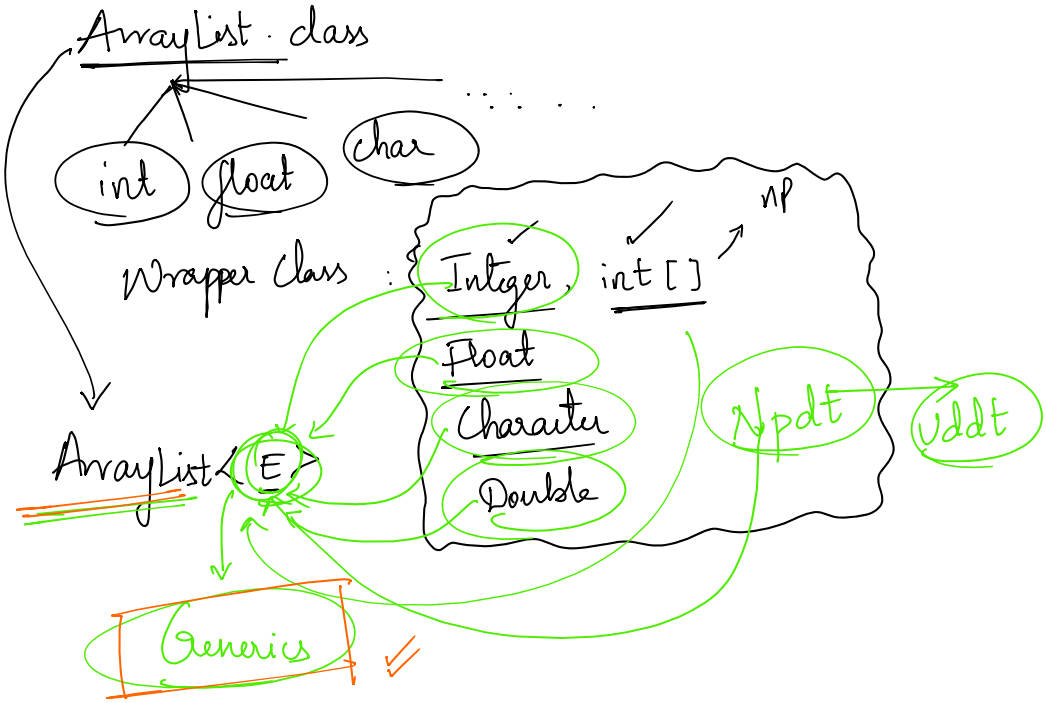
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Collections Framework

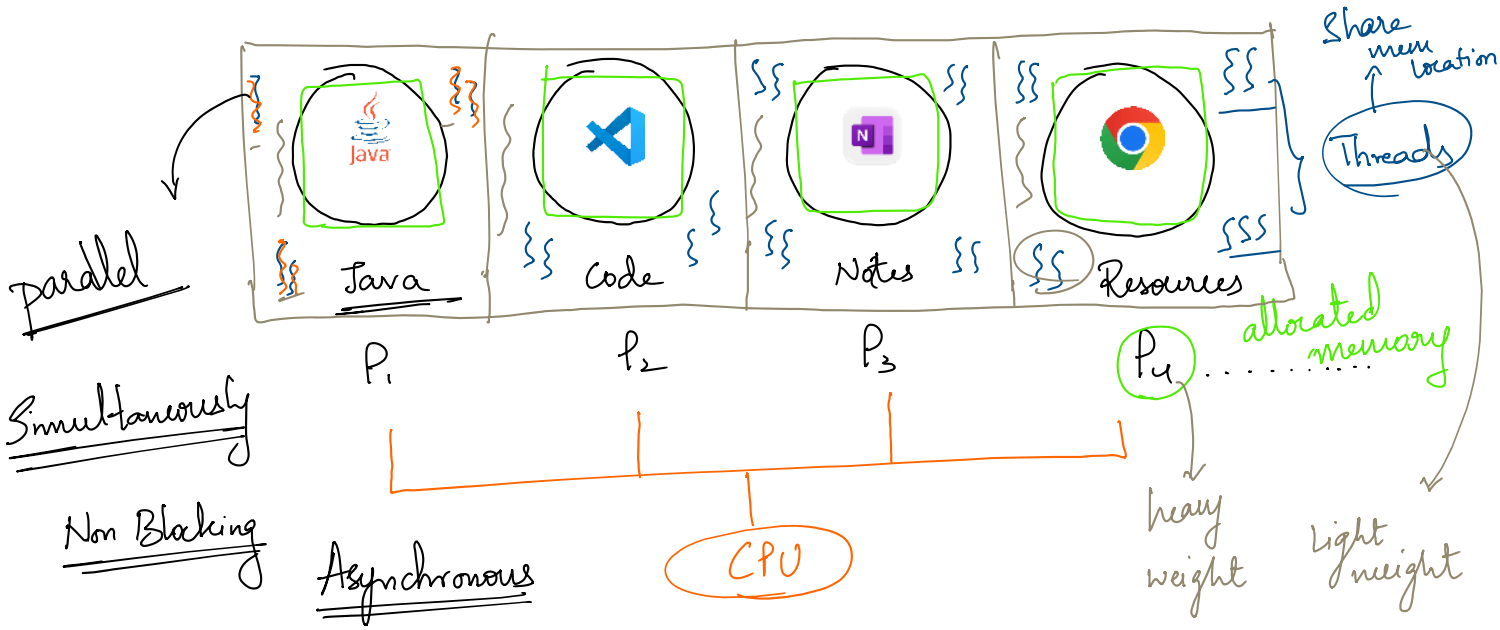
Monday, January 19, 2026 7:02 PM

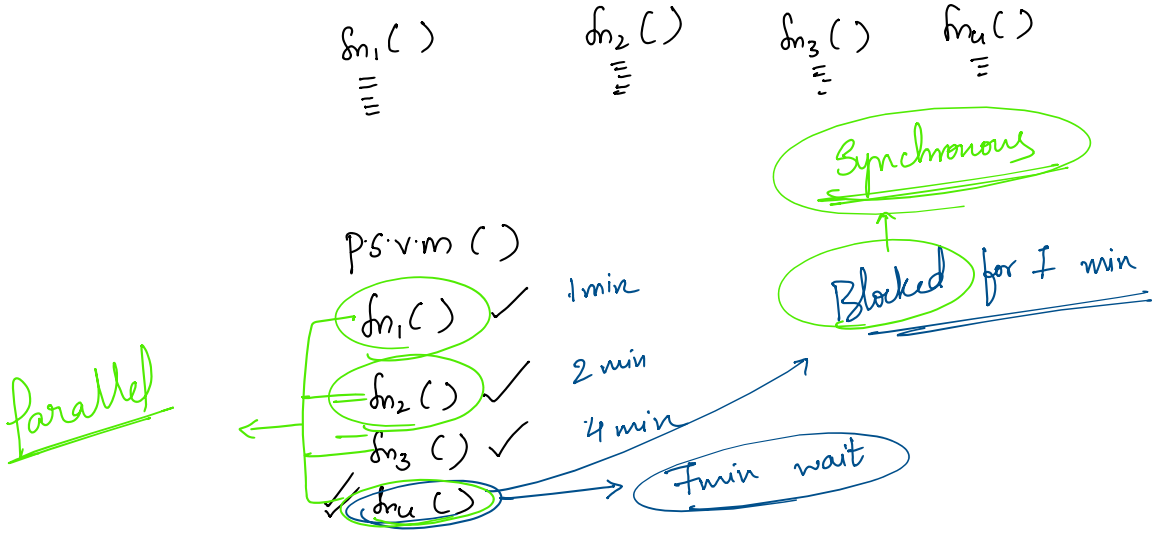


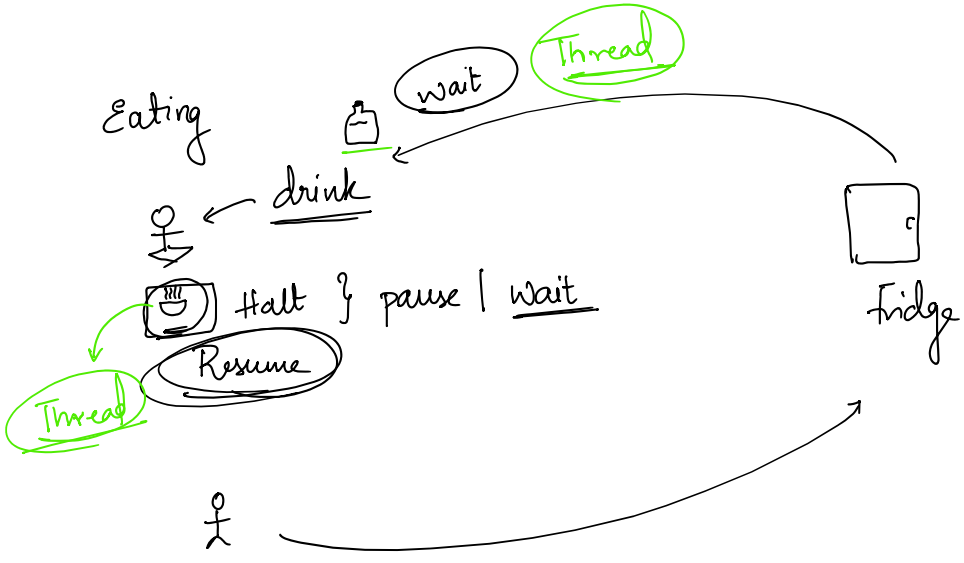


Processes vs Threads

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Thread Lifecycle

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